

# PRODUCTION SCIENTIFIQUE

Le cœur de ma recherche est le développement de techniques haut-contraste innovantes, essentielles pour détecter et caractériser des objets orbitant à faible séparation d'étoiles très brillantes (exoplanètes et ceintures de poussière). Au fil des années, j'ai acquis une expertise unique dans la compréhension analytique et la simulation numérique de ces instruments, à la fois au sol et dans l'espace, atteignant des performances sans précédent sur des bancs expérimentaux et directement en améliorant des instruments existants. En parallèle, je suis spécialiste de l'étude des exosystèmes, ayant analysé des données provenant de plusieurs instruments haut-contraste. Mes recherches intègrent de manière complémentaire le développement instrumental et l'analyse d'observations. À mesure que les instruments coronagraphiques deviennent plus complexes, maximiser leur potentiel nécessite une expertise en instrumentation, et les enseignements tirés des observations sur le ciel sont essentiels pour orienter les futurs développements instrumentaux.

**62 articles acceptés dans des revues à comité de lecture, dont :**

- **7 articles en tant que premier auteur + 1 en tant que co-premier auteur,**
- **10 articles avec contributions majeures (2ème ou 3ème auteur),**
- **44 articles avec contributions moins importantes.**

**53 actes de conférences (principalement SPIE), dont 11 en tant que premier auteur.**  
**Tous mes articles et actes sont en libre accès.**

**Google Scholar analytics : 3977 citations — h-index = 35.**

## PRINCIPAUX ARTICLES

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1. Squicciarini, V. ; **Mazoyer, J.** ; Wilkinson, C. et al. (2025), *GPI+SPHERE detection of a 6.1  $M_{Jup}$  circumbinary planet around HD 143811*, Astronomy and Astrophysics, 702, L10, [DOI link](#), [arXiv link](#), 7 citations
2. Stasevic, S. ; Milli, J. ; **Mazoyer, J.** et al. (2025), *Optimising reference library selection for reference-star differential imaging of discs with SPHERE/IRDIS*, Astronomy and Astrophysics, 701, A93, [DOI link](#), [arXiv link](#)
3. Leginja, I. ; Baudoz, P. ; **Mazoyer, J.** et al. (2025), *Extended linearity in the high-order wavefront sensor for the Roman Coronagraph*, Astronomy and Astrophysics, 698, A130, [DOI link](#), [arXiv link](#), 5 citations
4. Squicciarini, V. ; **Mazoyer, J.** ; Lagrange, A. -M. et al. (2025), *The COBREX archival survey: Improved constraints on the occurrence rate of wide-orbit substellar companions: I. A uniform re-analysis of 400 stars from the GPIES survey*, Astronomy and Astrophysics, 693, A54, [DOI link](#), [arXiv link](#), 13 citations
5. Gutierrez, Y. ; **Mazoyer, J.** ; Mugnier, L. M. et al. (2024), *Image-based wavefront correction using model-free reinforcement learning*, Optics Express, 32, 31247, [DOI link](#), [arXiv link](#), 2 citations
6. Galicher, R. ; Potier, A. ; **Mazoyer, J.** et al. (2024), *Increasing the raw contrast of VLT/SPHERE with the dark hole technique. III. Broadband reference differential imaging of HR4796 using a four-quadrant phase mask*, Astronomy and Astrophysics, 686, A54, [DOI link](#), [arXiv link](#), 6 citations
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10. Chen, C. ; **Mazoyer, J.** ; Poteet, C. A. et al. (2020), *Multiband GPI Imaging of the HR 4796A Debris Disk*, The Astrophysical Journal, 898, 55, [DOI link](#), [arXiv link](#), 42 citations
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12. **Mazoyer, J.** ; Pueyo, L. ; N'Diaye, M. et al. (2018), *Active Correction of Aperture Discontinuities-Optimized Stroke Minimization. I. A New Adaptive Interaction Matrix Algorithm*, The Astronomical Journal, 155, 7, [DOI link](#), [arXiv link](#), 18 citations

13. Fogarty, K. ; Pueyo, L. ; **Mazoyer, J.** et al. (2017), *Polynomial Apodizers for Centrally Obscured Vortex Coronagraphs*, The Astronomical Journal, 154, 240, [DOI link](#), [arXiv link](#), 10 citations
14. **Mazoyer, J.** ; Pueyo, L. ; Norman, C. et al. (2016), *Active compensation of aperture discontinuities for WFIRST-AFTA: analytical and numerical comparison of propagation methods and preliminary results with a WFIRST-AFTA-like pupil*, Journal of Astronomical Telescopes, Instruments, and Systems, 2, 011008, [DOI link](#), [arXiv link](#), 9 citations
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16. **Mazoyer, J.** ; Boccaletti, A. ; Augereau, J. -C. et al. (2014), *Is the HD 15115 inner disk really asymmetrical?*, Astronomy and Astrophysics, 569, A29, [DOI link](#), [arXiv link](#), 35 citations
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1. Landman, R. ; Doelman, D. ; Rietjens, J. et al. (2026), *SUPPPRESS: Prototyping and testing liquid-crystal vector vortex coronagraphs with reduced polarization leakage*, accepted in JATIS, [arXiv link](#)
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6. Desgrange, C. ; Milli, J. ; Chauvin, G. et al. (2025), *Dust populations from 30 to 1000 au in the debris disk of HD 120326: Panchromatic view with VLT/SPHERE, ALMA, and HST/STIS*, Astronomy and Astrophysics, 698, A183, [DOI link](#), [arXiv link](#), 5 citations
7. Chomez, A. ; Delorme, P. ; Lagrange, A. -M. et al. (2025), *The SPHERE infrared survey for exoplanets (SHINE): IV. Complete observations, data reduction and analysis, detection performances, and final results*, Astronomy and Astrophysics, 697, A99, [DOI link](#), [arXiv link](#), 12 citations
8. Ray, S. ; Sallum, S. ; Hinkley, S. et al. (2025), *The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems. III. Aperture Masking Interferometric Observations of the Star HIP 65426 at 3.8  $\mu$ m*, The Astrophysical Journal Letters, 983, L25, [DOI link](#), [arXiv link](#), 11 citations
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14. Hom, J. ; Patience, J. ; Chen, C. H. et al. (2024), *A uniform analysis of debris discs with the Gemini Planet Imager II: constraints on dust density distribution using empirically informed scattering phase functions*, Monthly Notices of the Royal Astronomical Society, 528, 6959, [DOI link](#), [arXiv link](#), 12 citations
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16. Worthen, K. ; Chen, C. H. ; Brittain, S. D. et al. (2024), *Vertical Structure of Gas and Dust in Four Debris Disks*, The Astrophysical Journal, 962, 166, [DOI link](#), [arXiv link](#), 6 citations
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36. Esposito, T. M. ; Duchêne, G. ; Kalas, P. et al. (2018), *Direct Imaging of the HD 35841 Debris Disk: A Polarized Dust Ring from Gemini Planet Imager and an Outer Halo from HST/STIS*, The Astronomical Journal, 156, 47, [DOI link](#), [arXiv link](#), 35 citations
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4. **Mazoyer, J.** ; Goulas, C. ; Vidal, F. et al. (2024), *Upgrading SPHERE with the second stage AO system SAXO+: non-common path aberrations estimation and correction*, Ground-based and Airborne Instrumentation for Astronomy X, 13096, 130969D, [DOI link](#), [arXiv link](#), 2 citations
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